Understanding Z motion

A short guide to what the model is showing

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Three things are easy to mix together

The model has changing internal complex states. It can display them as arrows on three fixed face planes. It also has a separate readout called Z, which combines information from the state and an observer's settings, then places a point in the fixed surrounding geometry.

That distinction is the useful result of this work. The point can change without the shell moving. It can change because the internal state changes, because the observer changes, or both. Rotating a camera is different again: it changes the view rather than the calculated point.

The author's fixed-support and overlapping-arrangement ideas prompted the investigation. Earlier cone and average-arrow interpretations were separate comparisons. These Z studies did not implement a moving support or establish a physical cone mechanism. The golden-ratio wording was illustrative, not a requirement to invent a controller.

What “frozen observer” means

The observer's angle and envelope time stay at their first-sample settings. The saved internal states still differ. A scheduled observer instead uses its declared angle and envelope settings at each saved index.

Frozen and scheduled points at one index come from the same state. They are not two successive native positions. Their difference reveals how the observer affects this particular readout.

There are two histories. The first has two saved states. The second has five, uses different native coefficients and a different starting state, and is not an extension of the first. The third study analyzes the same five-state history more deeply. Reusing it does not create another independent experiment.

A nearest opening can stay the same while the point moves

In the five-state history, frozen Z remains nearest the same upper boundary, G+0, at all five samples. Nevertheless, its position changes each time. The scheduled comparison changes nearest labels three times. In the stated accounting order, those label changes happen at the observer's clock replacement.

“Nearest” is only a comparison of distances. It does not mean the point is heading toward that opening or that energy is leaving there. In one frozen step, the point moves closer to the center while farther from its nearest boundary.

All primary points are inside the original reference sphere. That sphere is a measuring reference, not a demonstrated physical wall or material core. A ray from the center through one five-state point meets a separate notch triangle; the point itself does not. A direction test is not observed passage or emission.

Why the frozen point moves inward, then outward

Z blends two existing ingredients: a norm-dependent vector along a fixed direction and half the raw chirality vector. Chirality here is a mathematical cross product of the state's real and imaginary parts. The ingredients can reinforce in one coordinate and cancel in another.

All four saved steps matter

- 0→1: a large inward shift. Several large signed intensity terms oppose one another. The finite-step square term is essential; leaving it out would produce an impossible negative final intensity in that incomplete account.

- 1→2: a smaller inward shift. Two x contributions nearly cancel, even though y and height still change appreciably. A small x change does not mean nothing happened.

- 2→3: native intensity increases, yet the placed radius decreases slightly. The spatial blend outweighs the macro-size increase. This is a particularly clear example of why native intensity and distance from the center are different quantities.

- 3→4: the radius increases by about 0.00780087 width units, while height decreases slightly. Macro growth exceeds two opposing terms in the squared-radius balance. “Outward” does not mean “upward.”

There is no single term that plays the same geometric role in every direction. In the final step, one phase-related term contributes positively to y but negatively to the radial balance. The contribution sizes are not percentages of independent physical causes.

What this explains, and what it leaves open

The existing equations account for these retained changes without adding a clock, force or fitted movement law. Freezing the observer makes one comparison clearer; it does not turn the point into an observer-independent physical position.

The readout also discards some information. Giving every complex state component the same phase change leaves fixed staged Z unchanged, although the face arrows can change. Agreement of intensity and chirality budgets therefore cannot certify every component of a native state. EMA is a different existing observer with memory; no EMA calculation was part of these studies.

The results are numerical evidence from short, specific histories, not a proof of a continuous orbit, equilibrium, energy transport or gravity. The five-state example was deliberately constructed in earlier work. No general population or long-term behavior is inferred from it.

The methods note supplies the equations, both radial accounts, the exact source route and the numerical limitations. This guide and note only consolidate the completed evidence: no scientific calculations, model changes, Git commands or publication were performed. The recommended next decision is where to adopt the documentation after review, not an automatic new experiment.